The Research Lab Commons

Phase 1 Report - By Team: The System Error

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1. System Context Brief

1.1.System boundary and context

SERC and Precog lab occupy the same floor and share a nearby cafeteria area. Students need reliable space and equipment for research, project work and everyday breaks. The central problem is how limited resources become accessible through affiliation, established claims and personal negotiation.

The working boundary includes desk and chair allocation, local GPU access, food amenities (coffee machine, refrigerator), storage space, meeting rooms and the people who authorise or negotiate their use. Institute rules, maintenance and cleaning services, procurement and administrators enter as dependencies. Campus construction, overall admissions and research evaluation are outside direct control but shape capacity and demand.

1.2.Purpose and intended outcomes

The intended purpose is to support productive research and a usable shared workplace. Fairness means a transparent route to access based on legitimate need or urgency, continuity requirements and eligibility, with proportionate care duties. It does not require every resource to be freely available to every person. Relevant outcomes are fewer blocked work sessions, safe temporary desk use, predictable service and a route for quieter students to raise unmet needs.

1.3.Symptoms and initial problem framing

Participants describe more students than desks, names and belongings that signal enduring claims, GPU contention, and food waste left at desks. Several also describe cooperative borrowing and peaceful negotiation. The analytical question is therefore why a system that works through courtesy for established members can still leave newcomers uncertain or excluded.

Initial framing: How do resource boundaries, claim persistence, uneven information and distributed responsibility turn nominal capacity into unequal usable access?

The Phase 1 study maps existing arrangements and develops hypotheses for validation.

1.4.People and resource baseline

Element	Reported scale
Labs (5th floor)	SERC and Precog
Students	Total : about 100
Faculty	SERC: 7 ; Precog: 1
Desks and chairs	84
Local GPUs	
Food amenities	1 fridge, 1 coffee machine and 1 microwave
Meeting rooms and storage	2 meeting rooms locker count: 12

1.5.Capacity is conditional

A desk counts as usable only when it is physically workable, socially available and accessible at the required time. A space with no functioning light or ventilation was made usable after a student approached a faculty member. This is an example of recovering effective capacity without assuming that every empty surface already constitutes a usable desk.

The main constraints are fixed space, simultaneous demand, affiliation specific budgets, continuity needs for research setups, service response time and incomplete shared information.

1.6.System map

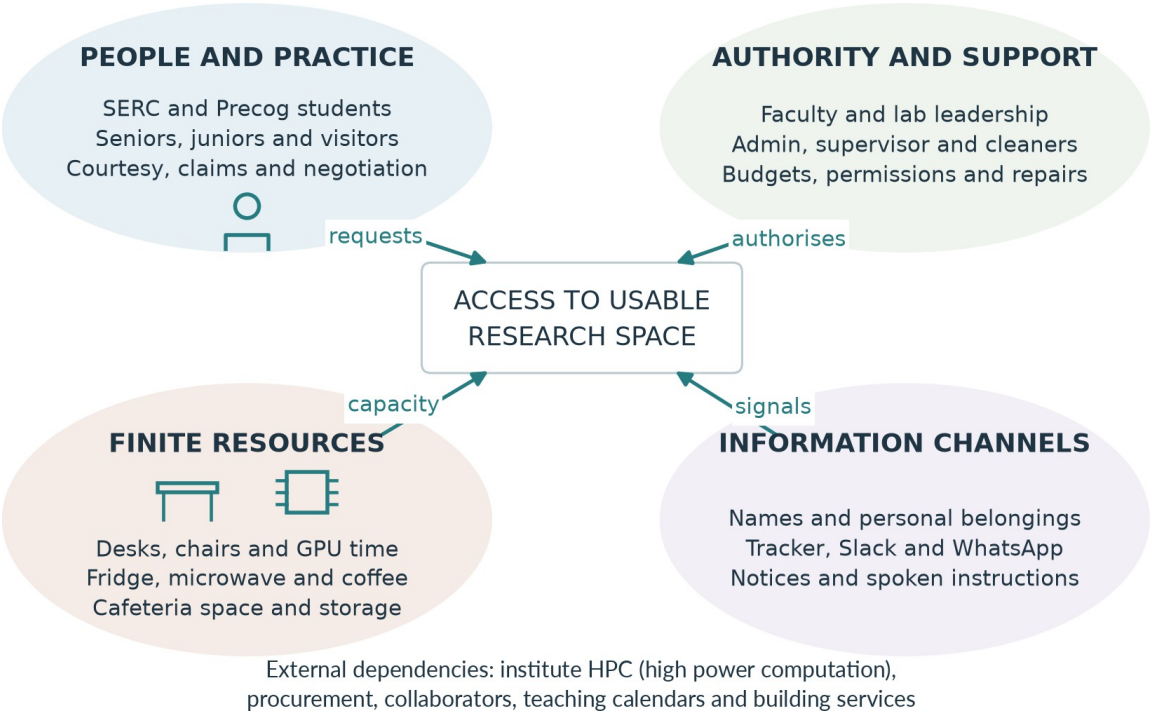


Figure 1. Clustered system map

Elements and interconnections

The resource layer supplies capacity. The authority layer sets eligibility and mobilises repair or purchasing. Information channels expose some claims and needs, while everyday peer practice translates those signals into actual access. The system has several local coordination arrangements rather than a single floor wide allocation process.

Operational purpose as an inference

The arrangements appear to protect continuity for recognised members and preserve sponsor control. That can coexist with helpful peer behaviour, yet it can leave temporary or less connected users carrying the burden of asking, waiting or moving. This is an inferred effect of the structure, not an allegation that individuals intend exclusion.

Boundary tension

The cafeteria is experienced as common space, but its appliances may be treated as belonging to a particular lab or eligible group. A shared location and a shared resource are different boundaries. Their mismatch explains why people can sincerely give incompatible accounts of fridge access.

2. Stakeholder Mapping

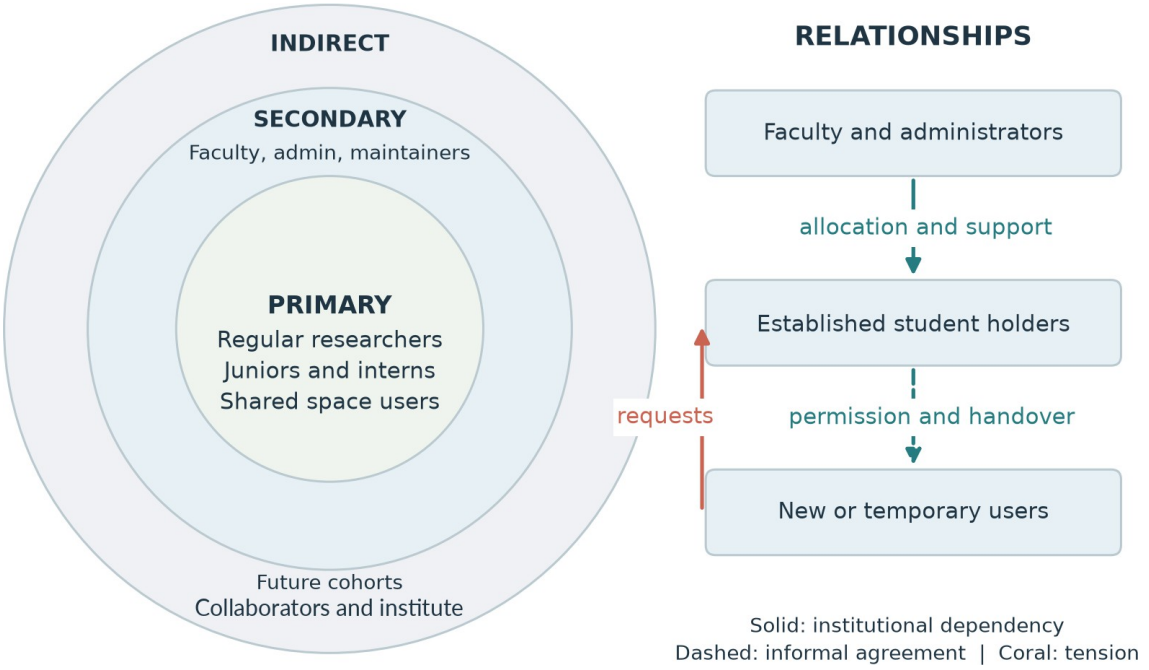


Figure 2. Stakeholder Map

Group and role	Interests and needs	Expectations and dependencies
Primary: established researchers	Stable setup, continuity, research progress	Expect a desk to remain available on return; depend on peers and faculty.
Primary: juniors, interns and temporary users	A usable place and a clear route to ask	Depend on vacancies, social confidence and holder consent; need predictable duration.
Secondary: faculty and resource sponsors	Research output and accountable resource use	Control funding or permissions; rely on student coordination.
Secondary: admin, supervisor, cleaners and student POCs	Serviceability and manageable workload	Need timely reports, stock and clear duties; do not necessarily set eligibility.
Indirect: institute, collaborators and future cohorts	Capacity stewardship and research continuity	Affected by local demand, collaboration needs and inherited allocation practices.

Formal and informal roles

Formal roles are defined by position and budget. Faculty and resource sponsors control funding and permissions, obtain collaborations, and can recover workspace or authorise repair. Administrators coordinate procurement, service requests and booking for the two meeting rooms, directly or through IMS. The supervisor and cleaning staff carry out upkeep, restocking and the scheduled Saturday fridge clearing, and report shortages upward rather than deciding eligibility. Institute services supply HPC, building maintenance and the constraints created by construction. None of these roles constitutes a common adjudicator for an everyday desk dispute.

Informal roles are defined by recognition rather than appointment. Established holders exercise practical control over desks and GPU time through names, belongings and repeated use; they can consent to temporary use without owning the resource, and retain priority on return. Seniors act as brokers, since a junior in Precog approaches a senior before approaching faculty. Student points of contact maintain the SERC tracker, and Precog's Slack channel makes demand visible while peers settle priority. Graduating seniors hand desks on by request. WhatsApp groups absorb complaints that never reach faculty, which is why formal authority stays under-informed.

Conflicts between groups

Conflict	Parties	Structural cause
Continuity against	Established holders vs new, junior and	No release or expiry rule, so an absent holder's claim persists and an empty desk is not readable as available
Reservation against	GPU holders vs deadline-driven users	A valid long job and an inactive held slot look alike without status detail
Ownership against	SERC and Precog, students vs eligible	The cafeteria is shared space while its fridge and coffee machine are treated as lab property; eligibility and hygiene rules are conflated
Benefit against	Students vs faculty, guests and service staff	Coffee access was restricted after volunteer maintenance failed; benefits were immediate while maintenance costs were delayed and
Care against convenience	Cleaning staff vs desk users	TA evaluations displace lunch into the labs, moving the cleaning burden to desks; staff carry the consequence without authority over
Voice against	Juniors vs seniors and faculty	Raising a small issue is socially expensive, so juniors yield and dissatisfaction never enters any formal record

3. Power Interest and Leverage

3.1.Power and interest map

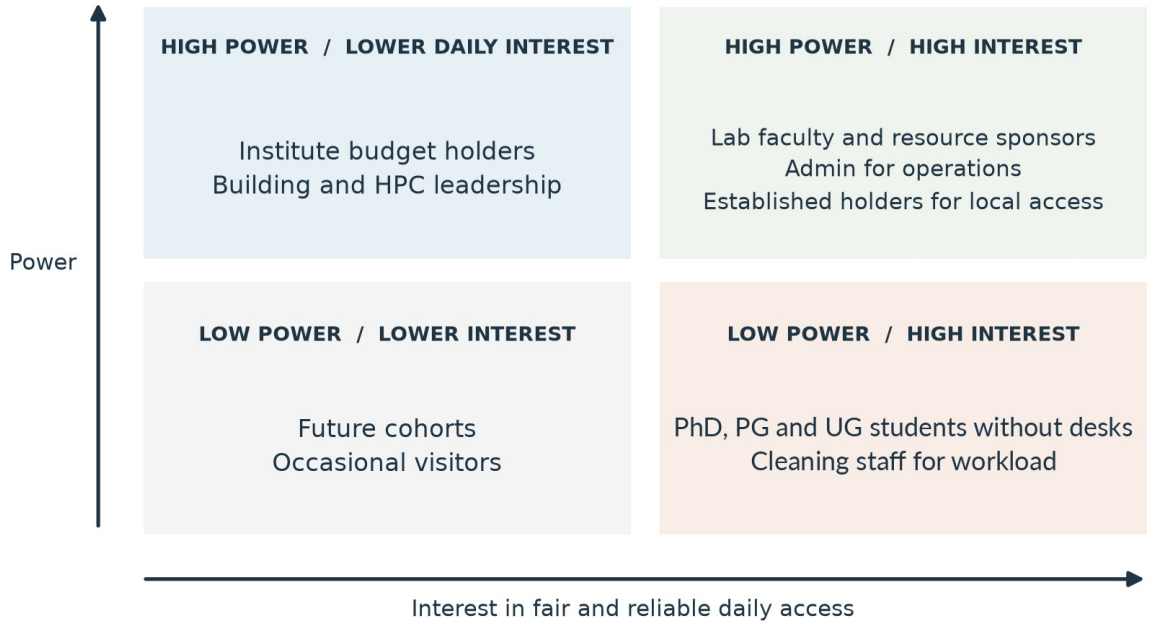


Figure 3. Qualitative power and interest grid

3.2.Formal and informal authority

Formal power lies with resource sponsors, faculty, administrators and institute services. Faculty can mobilise funding, obtain collaborations or recover workspace. Administrators coordinate service and room booking. These roles do not establish a common adjudicator for everyday desk disputes.

Informal power comes from a recognised claim, long tenure, familiarity with systems and access to helpful peers. A holder can consent to temporary use without formally owning the desk. A junior may avoid even asking, despite a senior saying borrowing is acceptable. This gap between offered permission and perceived permission is a mechanism of unequal access.

3.3.Incentives and unequal influence

Keeping a desk protects setup time and future certainty. Holding GPU time protects an experiment or deadline, while unused reservations shift delay onto others. Peer negotiation saves faculty attention but rewards those able to make persuasive requests. A person without an established claim cannot independently reallocate a desk or authorise access across a resource boundary.

Cleaning staff have high exposure to the consequences of use but limited authority over allocation. Low policy power must not be mistaken for low importance. Their workload and junior users' unmet needs should both be represented in decisions.

3.4.Leverage map

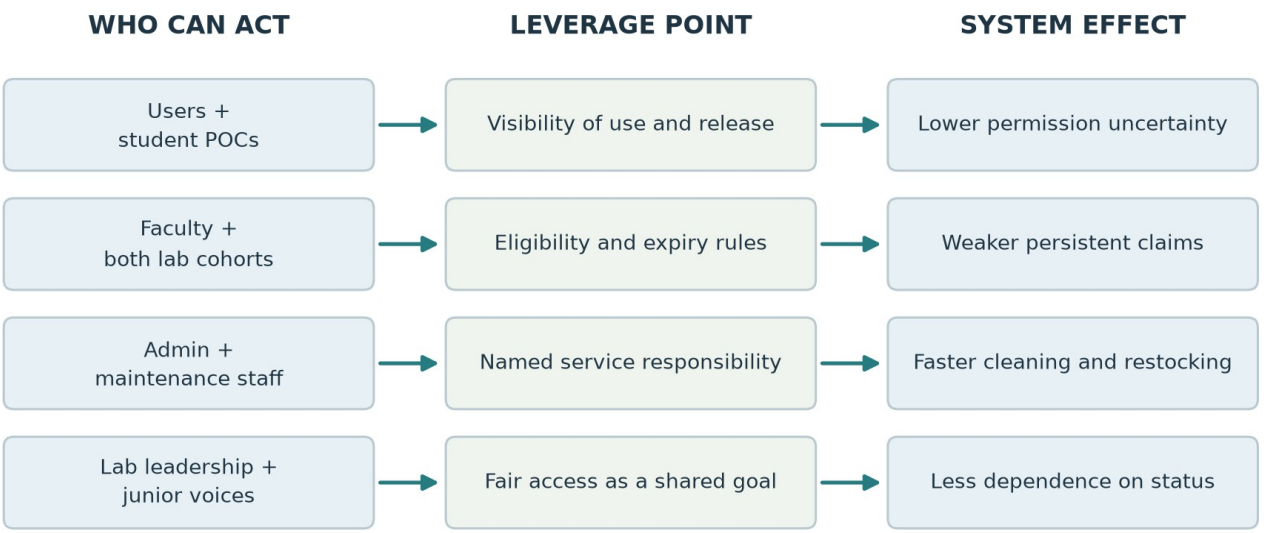


Figure 4. Actors linked to candidate leverage points and expected effects.

Candidate and mechanism	Constraint or unintended consequence	What would test it
Visible desk status and return date reduce uncertainty.	A register may legitimise indefinite claims unless expiry is defined.	Track idle but claimed desk hours and time needed to obtain permission.
Agreed temporary use and release rules weaken R1 (Reinforcing Loop 1)	Automatic release may disrupt long experiments, disability related needs or protected equipment.	Check temporary access success and continuity disruptions together.
Named care responsibilities strengthen B2 (Balancing Loop 2)	A student rota may shift unpaid work onto the most compliant people.	Record completion, staff workload and distribution of duties.
A joint forum gives juniors a route beyond favours.	A forum can become symbolic if it cannot change rules or allocate capacity.	Review whether raised needs receive decisions and whether both labs participate.

These are Phase 1 hypotheses, not approved interventions.

4. The Existing Process

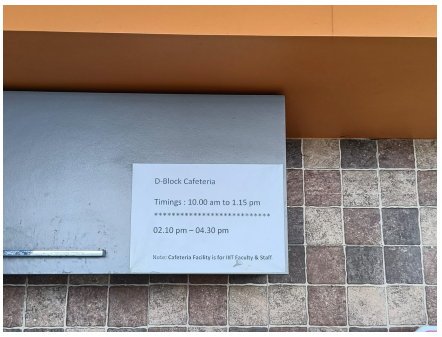
4.1.Research objectives and methods

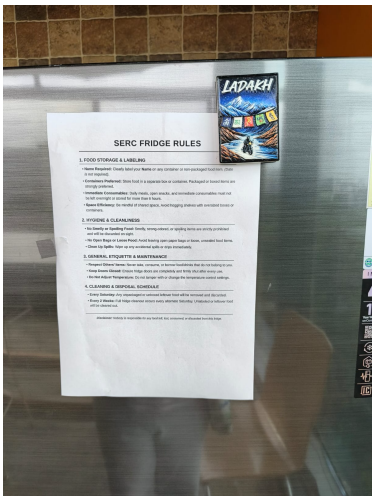
The enquiry asks who can use each resource, how a claim begins and ends, how urgent needs are negotiated, who restores depleted capacity, and what happens when agreement fails.

Analysis used thematic coding by resource, actor, rule, workaround, escalation and change over time. Accounts were compared for agreement and contradiction. First person experiences, secondhand reports and causal interpretations are distinguished.

4.2.Rules and actual practice

Resource	Reported rule or authority	Informal practice and exception
Desks	No common booking or release rule described.	Names, belongings and repeated use signal a claim. Borrowing may be allowed, but the holder has return priority. Graduating seniors pass desks through requests.
Local GPUs	SERC tracker and Precog Slack are separate arrangements.	Contact the user; negotiate urgency. Precog reports cross sponsor access through collaboration. Long unused reservations in SERC are secondhand claims.
Coffee Machine	Staff report new machine access for faculty and staff.	Earlier student access reportedly ended after upkeep failed. Some student accounts describe request based exceptions; these are not verified policy.
Fridge and microwave	Lab labelled fridge and Saturday cleaning notice reported.	Eligibility is contested; borrowing another member's name is described. Microwave fault is reported, without a verified repair route.
Meeting rooms	Admin checks availability for two rooms.	Direct admin contact or book through IMS







The field setting has pockets of explicit instruction, such as fridge notices, alongside unwritten allocation customs. "No written rules" is a useful starting concern, but not an accurate description of every resource.

4.3.Current state process map

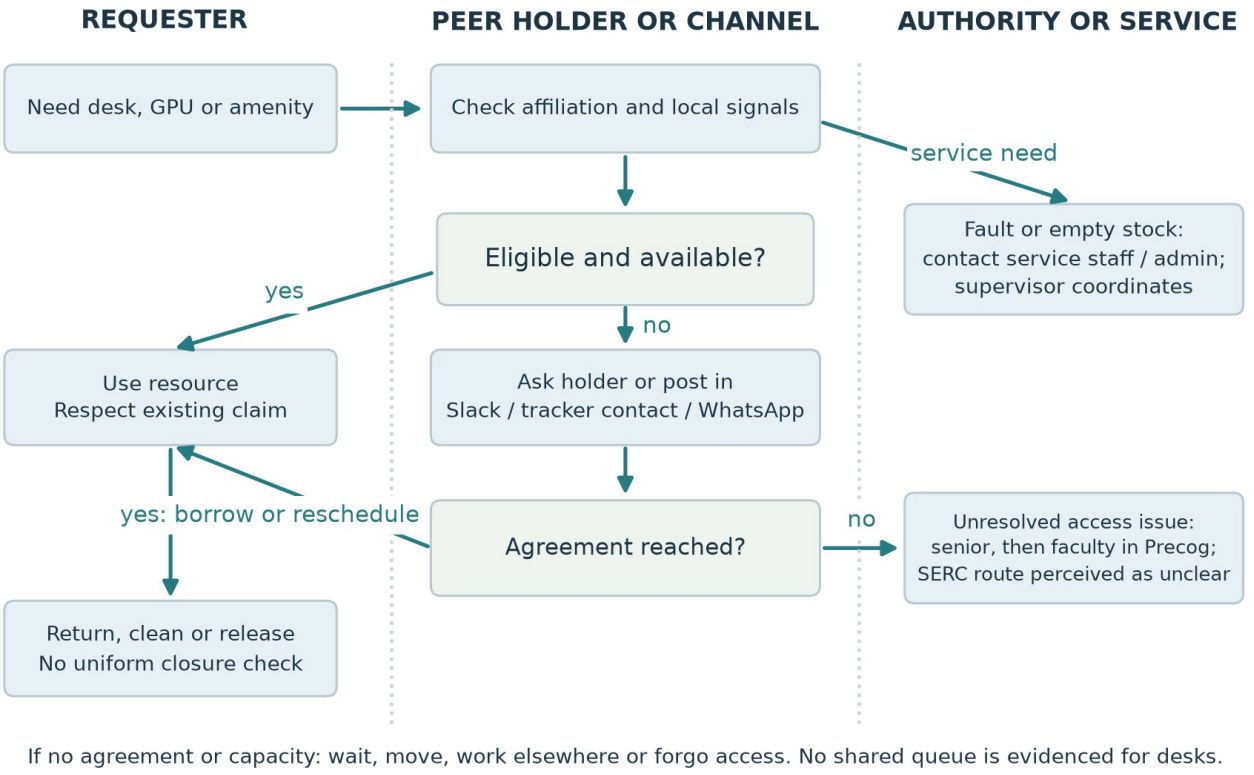
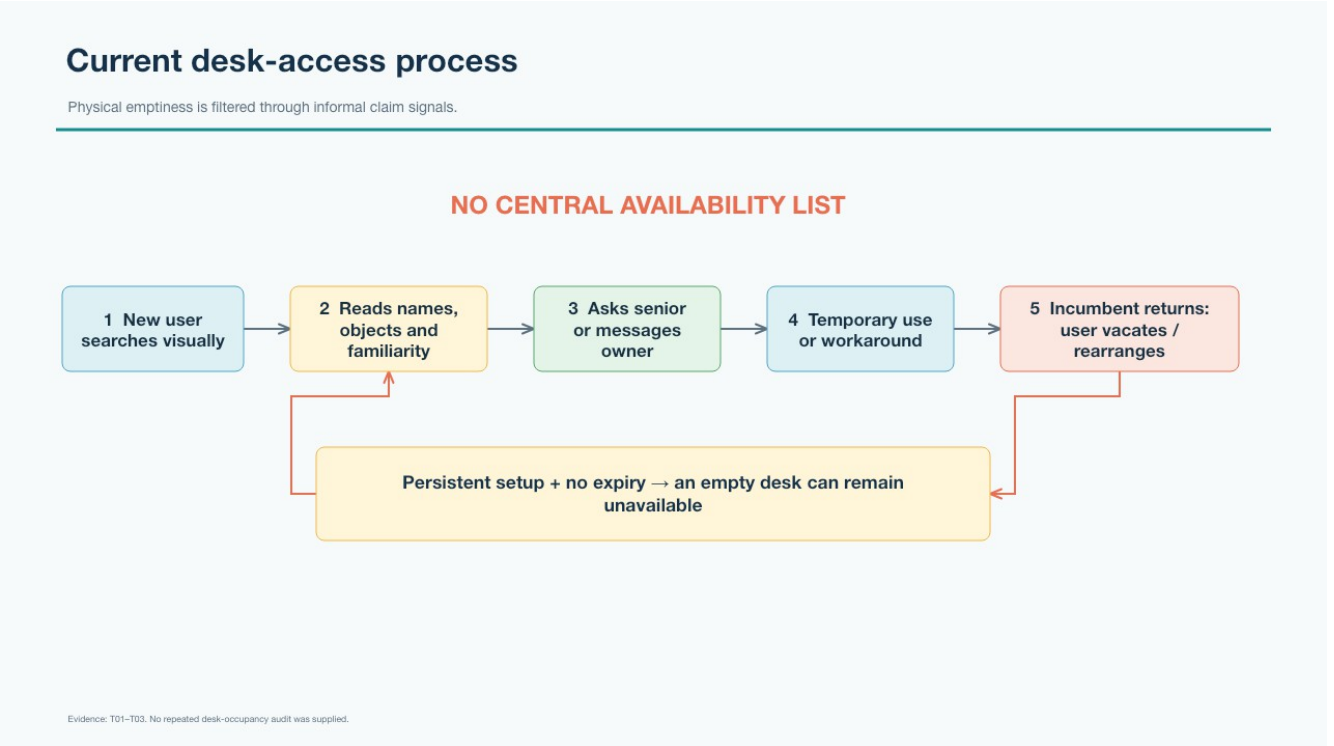
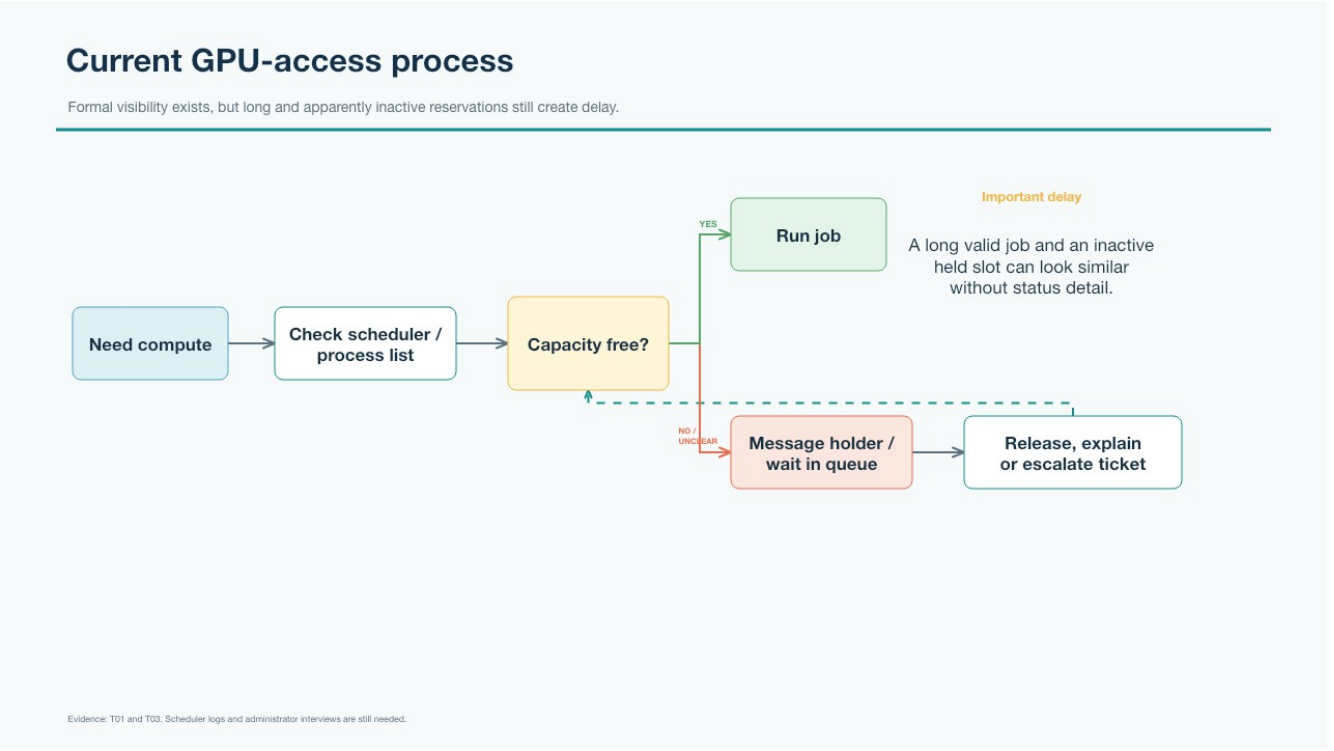


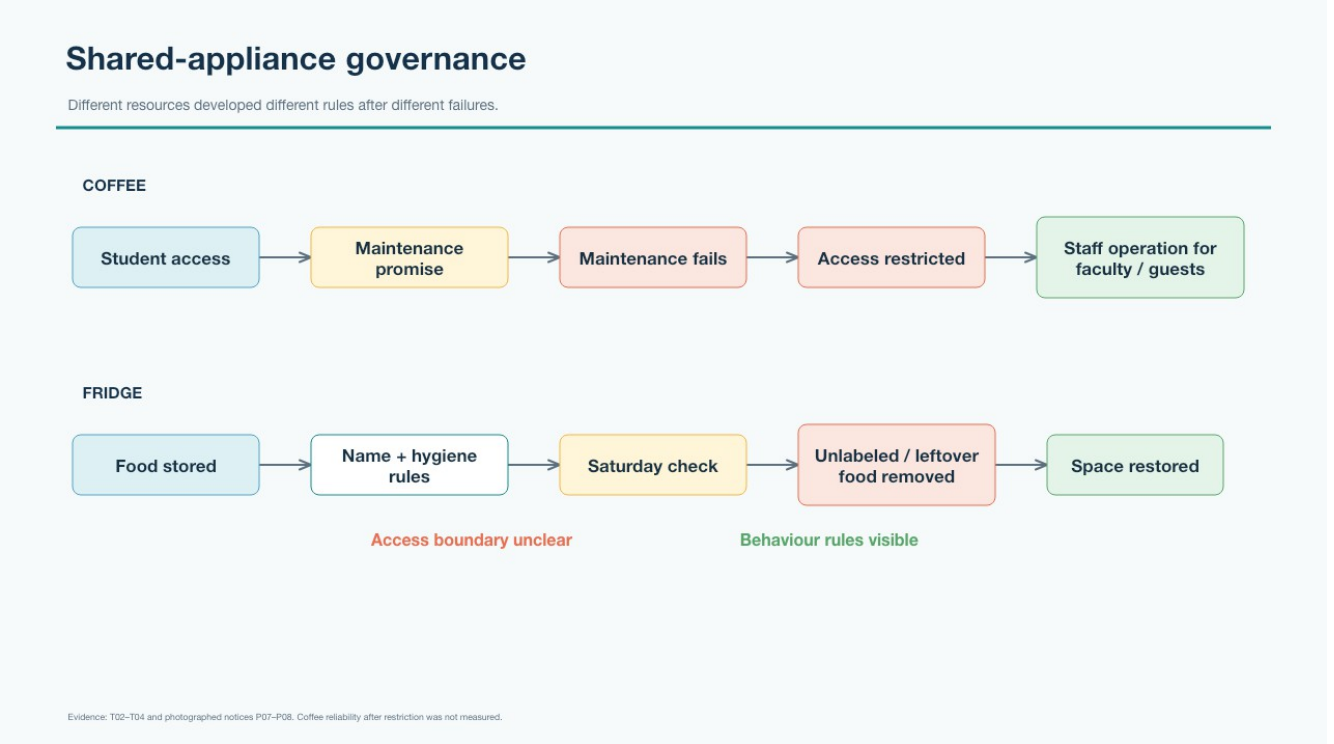
Figure 5. Current state swim lane trace. Decision boxes represent questions users resolve; routes reflect interview accounts, not a unified formal procedure.



A new user searches visually, interprets names and belongings, asks a senior or messages the established user, and either gains temporary use or creates a workaround. The established user retains priority on return. Because the system does not publish current availability or an absence threshold, physical emptiness is ambiguous.



GPU use has more formal information than desk use. Users can inspect running processes or the scheduler, contact the apparent holder, wait in a queue or raise a ticket. The process protects ongoing work but can delay urgent jobs when reservation status is unclear.



Coffee access reportedly moved from student use and a volunteer maintenance rotation to staff mediated faculty and guest use. Fridge behaviour is governed through names, hygiene rules and scheduled disposal. The different pathways show that governance emerged resource by resource, usually after a breakdown.





4.4.Dependencies and breakdowns

Access depends on knowing whose resource it is, finding the holder and securing agreement. SERC's tracker reduces uncertainty about GPU job ownership, but participants still contact the user before action. In Precog, Slack makes demand visible while peers determine priority. Neither visibility mechanism demonstrates that unused claims expire automatically. For service, staff report telling the admin department when coffee or milk is needed and informing a supervisor.

4.5.Negotiation and escalation

Precog describes a junior approaching seniors, then faculty if needed. SERC participants usually resolve matters among peers, with WhatsApp complaints and reluctance to involve faculty for small issues. Technical repairs have a clearer route to admin or faculty than interpersonal disputes. Silence is therefore not sufficient evidence that everyone is satisfied.

5. System Behaviour Over Time

5.1.Significant changes and triggers

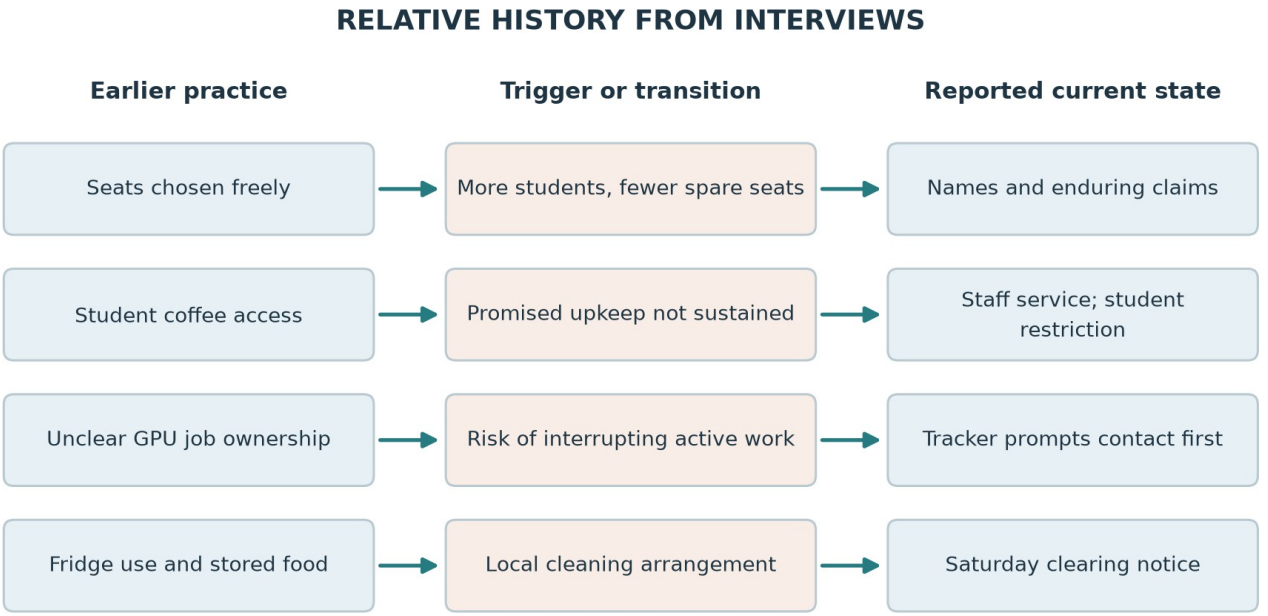


Figure 6. Relative timeline

5.2.Behaviour over time

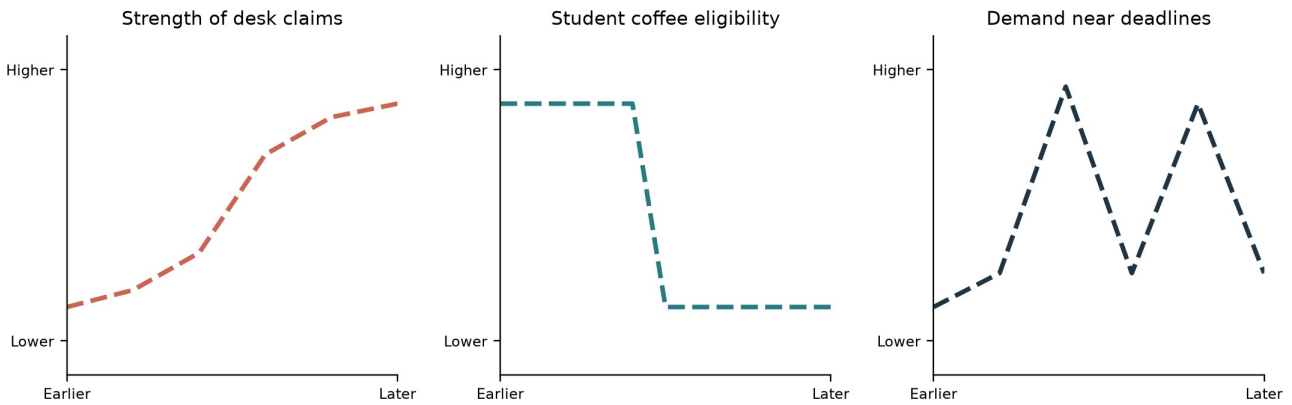


Figure 7. Dashed curves are qualitative hypotheses, not measured data.

The staff member says their cleaning of refrigerator involvement began about 15 days before the interview. One Precog participant recalls a roughly one week seating displacement.

5.3.What participants say and what the system does

Topic	Stated attitude	Reported or visible behaviour	Structural explanation
Sharing	People generally help when a need is genuine	New users still depend on seniors and incumbents	Help operates through unequal familiarity and private knowledge
Desk ownership	There is no formal personal ownership	Names, equipment and return expectations create durable possession	No release or expiry mechanism
Cleaning	Users should clean and maintain shared resources	Coffee maintenance reportedly failed and staff absorbed the work	Benefits are immediate while maintenance costs are delayed and distributed
Junior voice	Conflicts are peacefully discussed	Juniors may yield and small issues are not escalated	Formal feedback is socially expensive
Fridge access	Some describe the fridge as open to students	SERC labeling causes other users to perceive exclusion	Eligibility and behaviour rules are conflated

5.4.Iceberg model

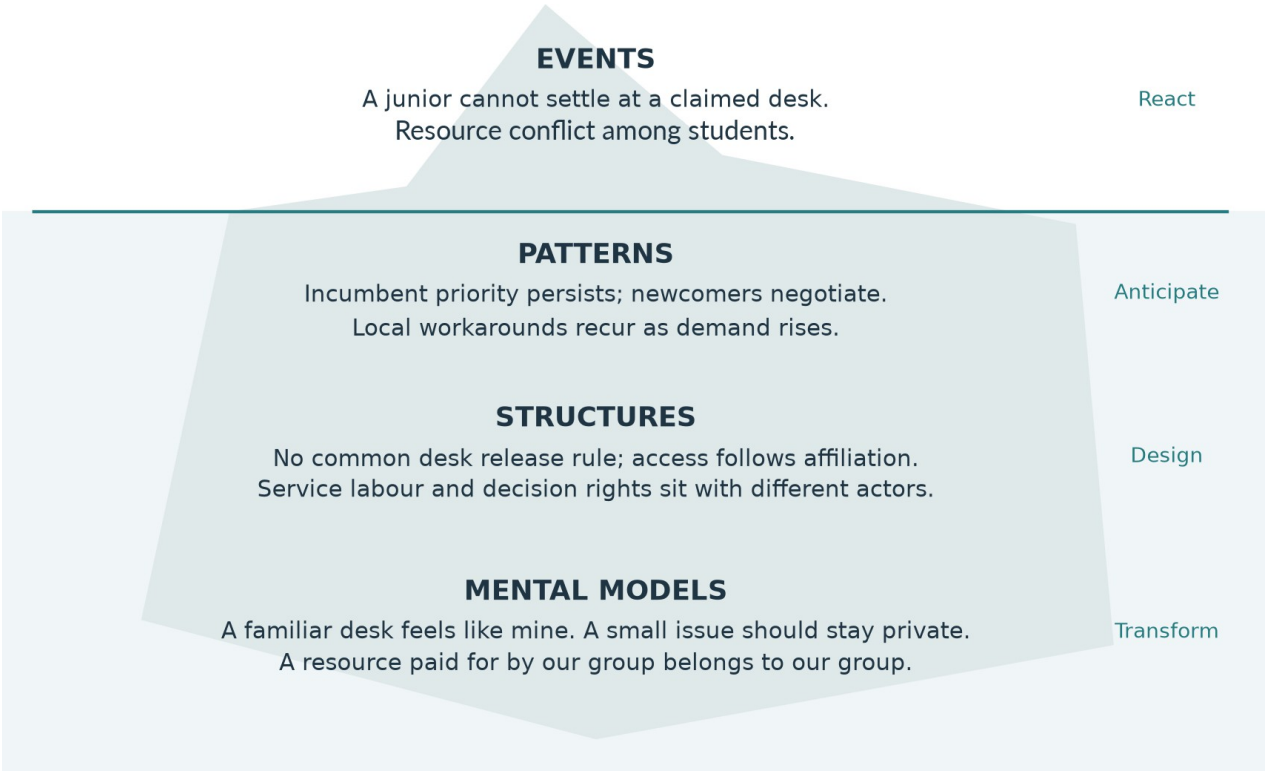


Figure 8. The Iceberg Model

From visible events to system structure

A newcomer's difficulty finding a desk is the visible event. Repeated incumbent priority is the pattern. Persistent claims without common release conditions form the structure. The belief that established use creates entitlement helps that structure survive, even when nobody intends to exclude a colleague.

Not all mental models are harmful. Respect for another person's equipment and a preference for peaceful dialogue support trust. The design challenge is to retain those benefits while reducing the need for personal permission and preventing small unresolved needs from disappearing from view.

A cross resource displacement mechanism

TA Evaluations can take chairs and occupy cafeteria space at lunch. A displaced student may eat at a work desk, shifting the cleaning burden into the lab. Restricting desk eating without restoring a place to eat could move the problem again. This is an interview supported causal chain, not a closed feedback loop.

5.5 Causal loop map

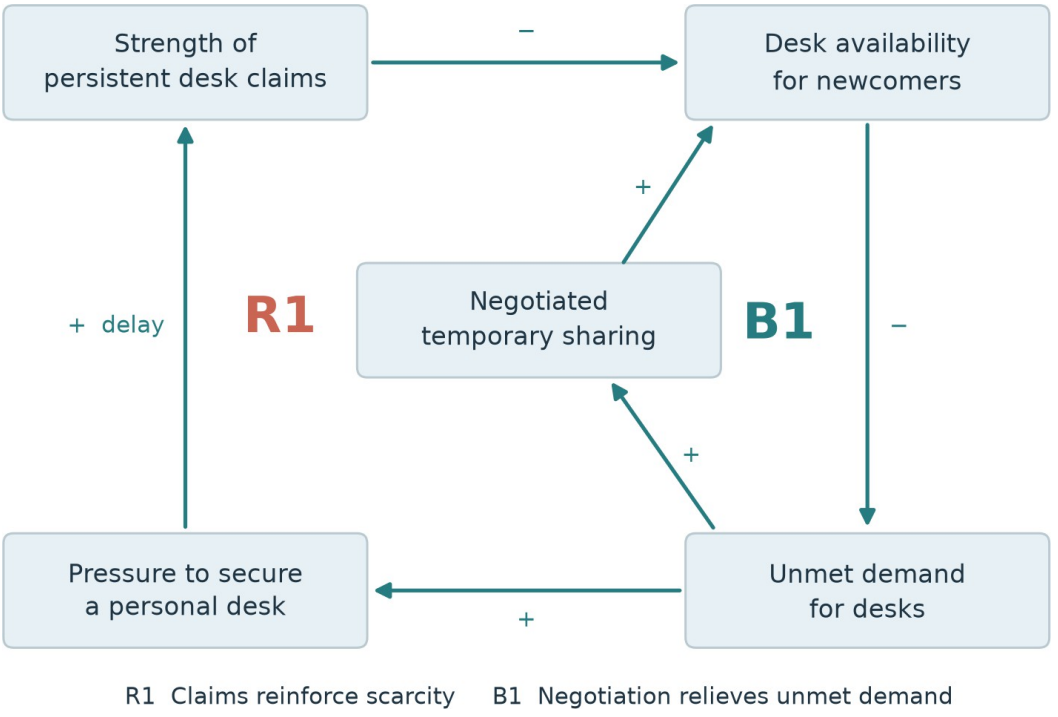


Figure 9. Coupled desk access loops. All arrows are causal hypotheses. R1 has two negative links and is reinforcing; B1 has one negative link and is balancing. A delay marks gradual establishment of a personal claim.

R1 Persistent claims reinforce scarcity

Stronger claims reduce newcomer availability. Lower availability raises unmet demand, which increases pressure to secure a personal desk. Over time, this pressure strengthens persistent claims. The first part is supported by borrowing and shortage accounts; the pressure to claim as a response is an inference requiring validation.

B1 Peer negotiation provides relief

Greater unmet demand prompts negotiated temporary sharing. Sharing increases availability and reduces unmet demand. This loop works when a holder is absent, reachable and willing, and when a requester feels able to ask. It cannot create space when all usable desks are occupied. A junior's hesitation can weaken the balancing response.

Stocks flows and delays

Stock and unit	Inflow and outflow	Why the distinction matters
Active desk claims, desks	New claims and renewals; releases and expiry	Absence is not a release. Furniture capacity can remain constant while accessible capacity changes.
Coffee and milk inventory, portions or volume	Restocking; consumption and spoilage	Empty cartons are a scenario symptom; interview stockout rates are not established.
Cleaning backlog, tasks or staff minutes	New mess; completed cleaning	A rota changes backlog only if completion matches incoming work.
Unserved desk demand, requests	New requests; fulfilled or withdrawn requests	Withdrawal lowers the queue but may conceal exclusion rather than indicate success.

Stocks capture the accumulations that give the system memory. Three conceptual stocks are particularly useful: genuinely available desks, GPU capacity available for new work, and trust in fair cooperation. Claims and reservations act as outflows from availability. Release, job completion, graduation, repair and added capacity act as inflows. Trust increases through transparent rules and helpful responses and decreases through opaque priority, mess and unresolved issues.

Important delays occur between absence and desk release, between a GPU reservation and actual use, between shortage detection and replenishment, and between a recurring minor problem and formal awareness. A delay can make a valid long experiment resemble an inactive reservation, or allow an obsolete desk claim to persist after the original need has ended.

5.6 Recurring patterns and unintended consequences

Claim persistence resembles commons enclosure: nominally shared capacity becomes difficult for others to access. The coffee history is consistent with a maintenance coordination failure followed by restricted access. It should not be labelled proof that students always over consume supplies. A fully demonstrated tragedy of the commons would require evidence of aggregate use exceeding replenishment or care capacity.

The tracker can reduce accidental interference yet leave long reservations untouched. Restricting coffee may reduce student related maintenance demand but also reduces student benefit. Quiet peer resolution can preserve relationships while concealing those who stop attending.

5.7 The system has no feedback loop yet

The system analysis describes change over time, but not learning. Nothing in the current arrangement routes information about unmet need back to the people who could act on it. A junior who gives up on a desk, a request that is quietly withdrawn and a fault that is never reported all leave no trace, so demand that disappears looks the same as demand that was met. Peer resolution keeps small problems inside WhatsApp groups and away from faculty, and no shared queue, occupancy record or closure check exists for desks. Governance has therefore moved only after visible breakdowns, as with coffee upkeep and fridge hygiene, rather than through a standing signal that reports how well access is working. Until such a return channel exists, the system cannot correct itself and the Phase 1 hypotheses cannot be tested against observed change.

6. System level conclusion

The Research Lab Commons is not only short of resources. It is short of trusted, current information about status, release, eligibility and responsibility. Under scarcity, users protect themselves through early claims, persistent reservations and social negotiation. These practices give incumbents continuity but transfer search, waiting and uncertainty to newcomers. Formal authority remains under informed because minor issues are filtered through peer networks. The outcome is an emergent system pattern that no individual actor needs to intend.